

HXC-108_video_qa_matrix

HXC-108 — Video-QA Recording Harness: Clients × Features Matrix

Owner: HXC-108 (video-QA recording harness) **Harness:** `scripts/video_qa/record_feature.sh` (+ `scripts/video_qa/find_window_id.swift`) **Revision:** 1 **Last modified:** 2026-06-22 **Status:** harness mechanics PROVEN (analyzer self-validation PASS); live window-scoped recording is an honest host TCC env-gap on the current dev host (§11.4.3) — see “Host capability gap” below.

This document enumerates HelixCode’s client surfaces × the features recordable per client, per §11.4.153 (per-feature status) / §11.4.158 (intensive recording coverage) / §11.4.143 (real-user-journey for app surfaces). It marks which features are recordable NOW via the CLI/TUI non-interactive paths versus which are GUI-input-dependent and therefore owned by the concurrent **HXC-112** desktop-GUI stream (this harness deliberately stays clear of HXC-112’s `helix_code/applications/desktop/`, `helix_code/scripts/testing/`, and any `helix_code-hxc112-*` recording — §11.4.119 single-resource-owner).

Coordination note. Two recording streams exist. HXC-112 drives the desktop Fyne GUI under `helix_code/scripts/testing/`. HXC-108 (this harness) lives under `scripts/video_qa/` and uses the recording scope-prefix `helixcode-<client>-<feature>-*` (self-test: `helixcode-harness_selftest-*`), so corpus rotation (§11.4.154(B)) only ever touches **this** harness’s own files.

Resolved facts (FACT, §11.4.6)

Fact	Value	Source
Project prefix (§11.4.151/.155)	helixcode	HELIX_RELEASE_PREFIX in <root>/ .env (env-first; dir-name fallback would be helix_code)
Recordings dir (§11.4.158 host override)	/Volumes/T7/Downloads/Recordings	root CLAUDE.md §11.4.158 project instantiation
Built client binaries	helix_code/bin/{helixcode, cli, tui/helix-tui, helix-desktop, helixcode-web-fresh, helixcode-infra}	ls helix_code/bin/
Client app sources	helix_code/applications/{desktop, terminal_ui,	ls helix_code/applications/

Fact	Value	Source
	android, ios, aurora_os, harmony_os}	

Clients × recordable features

Legend — **Input class:** CLI/TUI = scriptable non-interactive, recordable by THIS harness now · GUI = GUI-input-dependent, **HXC-112-gated** (not recorded here) · SERVER = HTTP/API, recordable via curl-driven terminal · MOBILE = device/emulator, future phase.

Client	Binary / source	Feature	Recordable command (deterministic)	Input class	Owner
CLI	helix_code/bin/cli	version banner	cli -version	CLI	HXC-108 no
CLI	helix_code/bin/cli	health check	cli -health	CLI	HXC-108 no
CLI	helix_code/bin/cli	list models (CONST-036, LLMsVerifier)	cli -list-models	CLI	HXC-108 no (needs verifier providers reachable)
CLI	helix_code/bin/cli	list workers	cli -list-workers	CLI	HXC-108 no
CLI	helix_code/bin/cli	non-interactive generate (real LLM)	cli -non-interactive -prompt "What is 2+2?" -model <m>	CLI	HXC-108 no (needs provider key Ollama)
CLI	helix_code/bin/cli	streaming generate	cli -non-interactive -stream -prompt "..."	CLI	HXC-108 no (provider-gated)
CLI	helix_code/bin/cli	command execution (BLUFF-003 path)	cli -command "echo hi"	CLI	HXC-108 no
CLI	helix_code/bin/cli	QA bank list / run	cli -qa-list · cli -qa-run -qa-banks 	CLI	HXC-108 no (qa-server gated for -qa-run)
TUI	helix_code/bin/tui (helix-tui)	terminal-UI boot + panes	launch in Terminal window, drive keys	TUI	HXC-108 no (window-scoped)

Client	Binary / source	Feature	Recordable command (deterministic)	Input class	Owner
Server	helix_code/bin/helixcode	HTTP boot + health endpoint	start server, curl / health in same window	SERVER	HXC-108 no
Server	helix_code/bin/helixcode	/api/v1/llm/generate real output	curl -X POST .../generate -d '{...}'	SERVER	HXC-108 no (infra+provi gated)
Desktop GUI	helix_code/applications/desktop (helix-desktop)	Fyne window, chat, settings, model picker	GUI click/type	GUI	HXC-112
Web	helix_code/bin/helixcode-web-fresh	browser UI	browser-driven (chromedp/ playwright)	GUI	HXC-112 / future
Mobile (iOS)	helix_code/applications/ios	app launch + chat journey (§11.4.143)	simulator UI journey	MOBILE	future phase
Mobile (Android)	helix_code/applications/android	app launch + chat journey (§11.4.143)	emulator UI journey	MOBILE	future phase
Mobile (Aurora/ Harmony)	helix_code/applications/{aurora_os,harmony_os}	app launch	device journey	MOBILE	future phase

Recording + validation contract (per feature)

For each recordable feature the harness produces, per §11.4.159: 1. **Window-scoped** capture only (screencapture -l<id> / -v -l<id>) — never whole-desktop (§11.4.154). 2. Output **MP4 H.264** +faststart yuv420p at /Volumes/T7/Downloads/Recordings/helixcode-<client>-<feature>-<ts>.mp4 (§11.4.155). 3. **Fresh-corpus rotation** of ONLY this harness's own scope-prefixed prior files first (§11.4.154(B)). 4. **Read-the-screen** OCR content-validation against --expect patterns, anti-bluff scan for TODO implement/ simulate/placeholder/for now (§11.4.158/.160/.163); PASS only when expected content is genuinely read back AND no bluff pattern present. 5. **Self-validated analyzer** (golden-good PASS + golden-bad FAIL) so the analyzer itself cannot bluff (§11.4.107(10)).

Host capability gap (§11.4.3 — FACT, captured, never faked)

record_feature.sh probe-caps on the current dev host (2026-06-22, macOS 15.5)
reports:

```
[OK ] screencapture -R region still
[GAP] screencapture -l<id> window still — "could not create image from w
[GAP] screencapture -v window video — "The operation could not be com
[OK ] OCR (magick render + tesseract read) under TMPDIR=/Volumes/T7/tmp
```

Interpretation: macOS TCC on this host grants Screen-Recording for region/full-screen STILLS, but blocks both window-by-id capture primitives (`-l <id> still` AND `-v video`). Because §11.4.154 forbids a whole-desktop/region substitute for a window, the harness SKIPS live recording with the exact failing primitive + reproducer rather than fake it. The harness is otherwise fully proven (see self-test evidence in `docs/qa/HXC-108_selftest_evidence.md`). To unblock live recording: grant the controlling app full Screen-Recording (incl. video) in System Settings → Privacy & Security → Screen Recording, then run from a GUI-foreground session — OR run the harness on a host where `probe-caps` shows [OK] for the window primitives.

Sources verified

- `helix_code/bin/cli -help` (flag inventory), `ls helix_code/bin`, `ls helix_code/applications` — 2026-06-22.
- Host capability probe `record_feature.sh probe-caps` — 2026-06-22.